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PREAMBLE

DO NOT REMOVE THE LINE BELOW

```
clear;
```

QUESTION 1: TEAM!

======================= Thomas Hiendel

QUESTION 2: COMMENTING

=======================

```
% Copy and comment every line of the following MATLAB script. Say what
% each line is doing in your comment. Explain each MATLAB line by using
% no more than one comment line, as done in the first line below. Run and
% publish the script: a=zeros(1,5); % Generate and print a 1x5 row
vector of zeros b=ones(3,2); % Generate a 3x2 grid of ones c=size(a); %
Finds the size of a, which is 1 5 abs([-5.2 , 3]); % Finds absolute of
-5.2, 3, which is 5.2000 3.0000 floor(3.6); % Rounds 3.6 down to 3
d=[1:-3.5:-9]; % Starts at 1 stepping in intervals of -3.5 to/before -9
f=d(2); g=sin(pi/2); % Sets f to -2.5, g to 1
K=[1.4, 2.3; 5.1, 7.8]; % Declares two row, two column 2D matrix
m=K(1,2); % Sets m to the first row, second column (2.3)
n=K(:,2); % Sets n to the second column, 2.3 and 7.8 comp =
3+4i; % Sets comp to the real and imaginary components
real(comp) % Takes the real component of comp (3) imag(comp) %
Takes the imaginary component of comp (4) abs(comp) % Takes the
absolute value of the comp (5) angle(comp) % Finds the angle of
comp (0.9273)
disp('haha, MATLAB is fun'); % Prints: haha, MATLAB is fun
3^2; % Squares 3 (9)
4==4; % Does a boolean check (True)
[2==8 3~=5]; % Does two boolean checks in a matrix [False True]
x=1:2:8; % Sets x to [1 3 5 7] y=[5 7 6 8]; % Sets y to [5 7 5
8]
```

```
q = zeros(10,1); % Sets q to 10 rows and one column of zeros for ii = 1:10
% Sets ii to 1, stepping up to 10 in intervals of 1      q(ii) = ii^2; %
Sets row ii to ii^2 end figure(1021); % Creates empty figure meeting
stem(x,y) % Makes a stem plot with our x and y values defined earlier hold
on; % Prevents new plots from clearing out old plots plot(x,y, 'k',
'linewidth', 2) % Draws a solid black line plot(x,y,'+r', 'markersize',
20) % Draws red signs at the coordinate pairs hold off; % Disables hold
above xlabel('Horizontal Axis') % Labels x axis as 'Horizontal Axis'
ylabel('Vertical Axis') % Labels y axis as 'Vertical Axis'
```

ans =

3

ans =

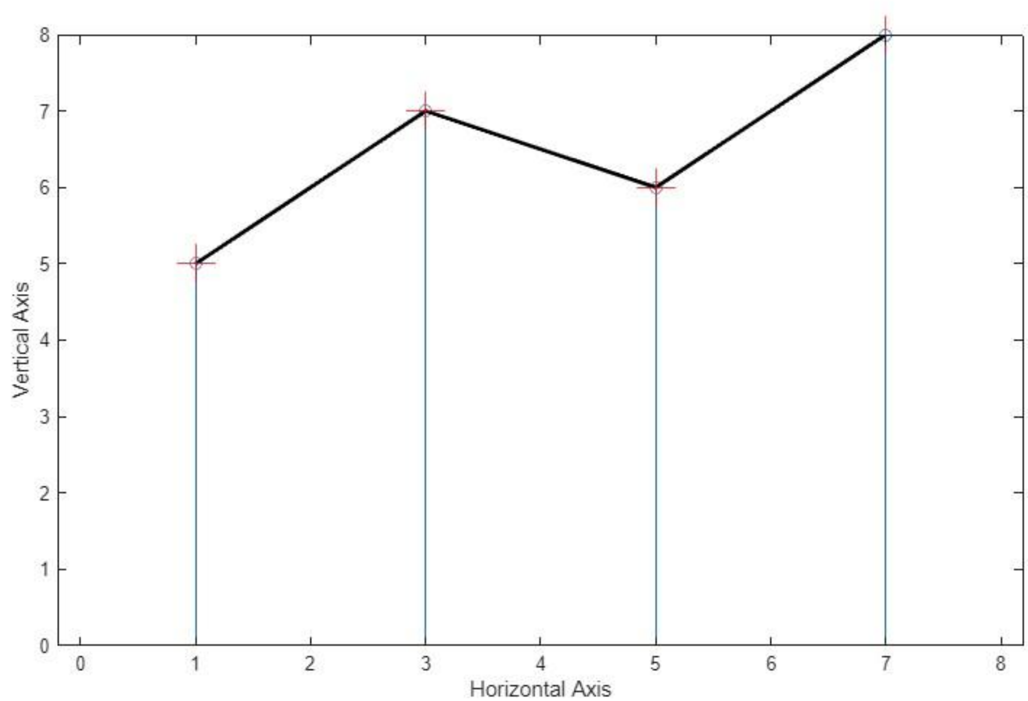
4

ans =

5

ans =   0.9273

haha, MATLAB is fun

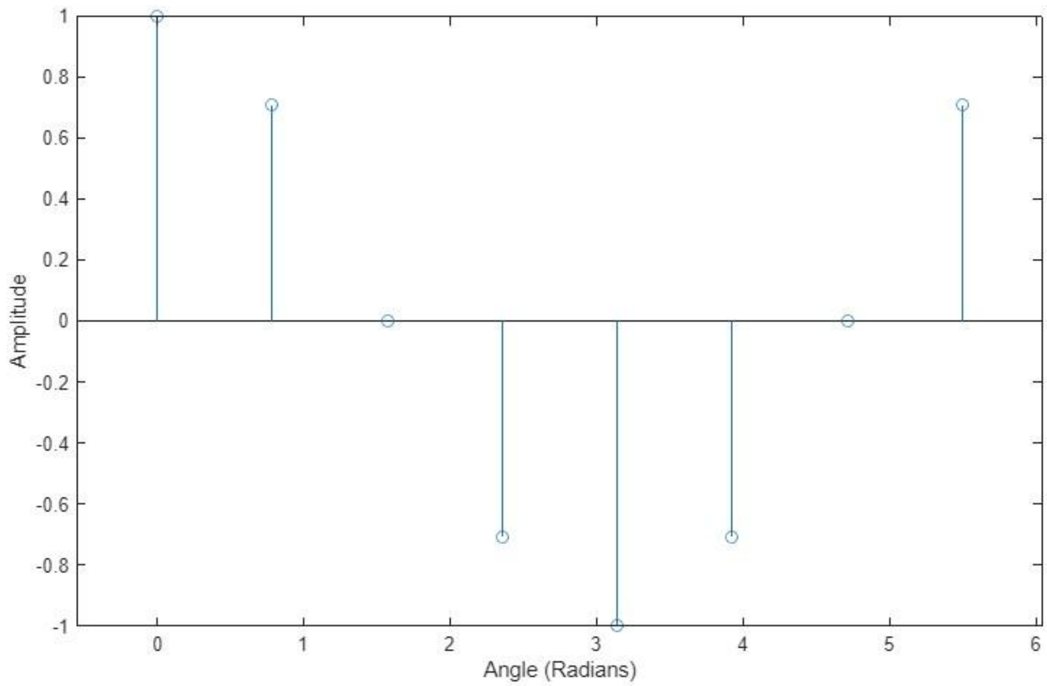


QUESTION 3: PLOTTING

========================

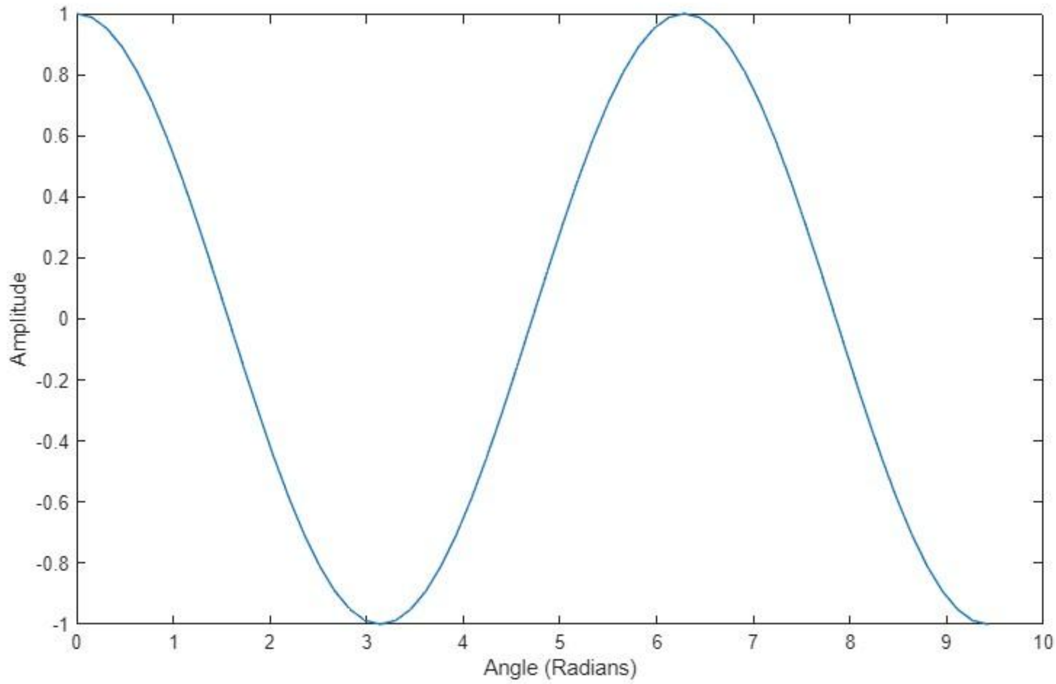
3(a) PLOT RESULT

```
figure(1) vect1 = [0 pi/4 2*pi/4 3*pi/4 4*pi/4 5*pi/4
6*pi/4 7*pi/4]; vect2 = cos(vect1); stem(vect1, vect2)
xlabel('Angle (Radians)') ylabel('Amplitude')
```



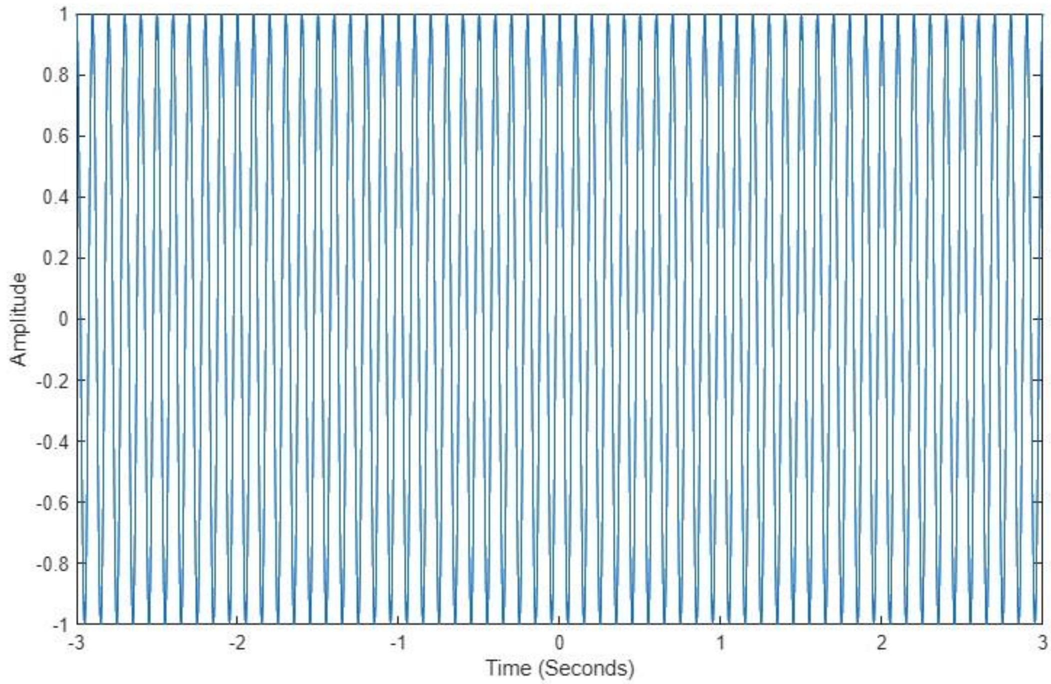
3(b) PLOT RESULT

```
figure(2) theta =
0:pi/20:3*pi; y =
cos(theta); plot(theta,
y) xlabel('Angle
(Radians)')
ylabel('Amplitude')
```



3(c) PLOT RESULT

```
figure(3); time = -
3:0.0001:3; y =
cos(20*pi*time);
plot(time, y)
xlabel('Time (Seconds)')
ylabel('Amplitude')
```



QUESTION 4: COMPLEX ROOTS

======================

4(a) WRITE FUNCTION IN SEPARATE FILE (TEMPLATE PROVIDED)

type('myroots.m')

```
% function r = myroots(n, a)
% myroots: Find all the nth roots of the complex number a
% % Input
Args:
% n: a positive integer specifying the nth roots
% a: a complex number whose nth roots are to be returned
% %
Output:
% r: 1xn vector containing all the nth roots of a

% A = abs(a); % Find the magnitude of the complex number
% phi = angle(a); % Find the phase (phi) of the complex number
% k = [0:1:n-1]; % Setting k to given value
% r = A^(1/n)*exp(j*(phi + 2*pi*k)/n) % Using root solve formula

% end
```

4(b) ANSWER QUESTION

```
help myroots % prints the documentation of the myroots function
```

```
  myroots: Find all the nth roots of the complex number a
  Input Args:   n: a positive integer specifying the nth
roots   a: a complex number whose nth roots are to be
returned
  Output:   r: 1xn vector containing all the nth
roots of a
```

4(c) OUTPUT RESULTS

```
r = myroots(9,2) r
= myroots(23,-j)
```

```
r =   Columns 1 through 4    1.0801 + 0.0000i   0.8274 + 0.6942i   0.1876 +

1.0637i  -0.5400 + 0.9354i

  Columns 5 through 8   -1.0149 + 0.3694i  -1.0149 - 0.3694i  -0.5400 -

0.9354i   0.1876 - 1.0637i

  Column 9

0.8274 - 0.6942i


r =   Columns 1 through 4    1.0801 + 0.0000i   0.8274 + 0.6942i   0.1876 +

1.0637i  -0.5400 + 0.9354i

  Columns 5 through 8   -1.0149 + 0.3694i  -1.0149 - 0.3694i  -0.5400 -

0.9354i   0.1876 - 1.0637i

  Column 9

0.8274 - 0.6942i


r =

  Columns 1 through 4
   0.9977 - 0.0682i   0.9791 + 0.2035i   0.8879 + 0.4601i   0.7308 + 0.6826i

  Columns 5 through 8    0.5196 + 0.8544i   0.2698 + 0.9629i   0.0000 +

1.0000i  -0.2698 + 0.9629i
```

Columns 9 through 12   -0.5196 + 0.8544i -0.7308 + 0.6826i -0.8879 + 0.4601i -0.9791 + 0.2035i

Columns 13 through 16   -0.9977 - 0.0682i -0.9423 - 0.3349i -0.8170 - 0.5767i -0.6311 - 0.7757i

Columns 17 through 20   -0.3984 - 0.9172i -0.1362 - 0.9907i   0.1362 - 0.9907i   0.3984 - 0.9172i

Columns 21 through 23   0.6311 - 0.7757i   0.8170 - 0.5767i   0.9423 - 0.3349i

r =   Columns 1 through 4   0.9977 - 0.0682i   0.9791 + 0.2035i   0.8879 + 0.4601i   0.7308 + 0.6826i

Columns 5 through 8   0.5196 + 0.8544i   0.2698 + 0.9629i   0.0000 + 1.0000i -0.2698 + 0.9629i

Columns 9 through 12   -0.5196 + 0.8544i -0.7308 + 0.6826i -0.8879 + 0.4601i -0.9791 + 0.2035i

Columns 13 through 16   -0.9977 - 0.0682i -0.9423 - 0.3349i -0.8170 - 0.5767i -0.6311 - 0.7757i

Columns 17 through 20   -0.3984 - 0.9172i -0.1362 - 0.9907i   0.1362 - 0.9907i   0.3984 - 0.9172i

Columns 21 through 23   0.6311 - 0.7757i   0.8170 - 0.5767i   0.9423 - 0.3349i

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